Lectures 12 – 14

Math 4741: Mathematical Analysis

Chapter 7: Queueing Theory

Texts:

Markov Processes for Stochastic Modeling – Oliver Ibe (Chapter 7)

Objectives

1. Understanding Markovian queueing systems
2. Derive steady-state analysis of Markovian queues

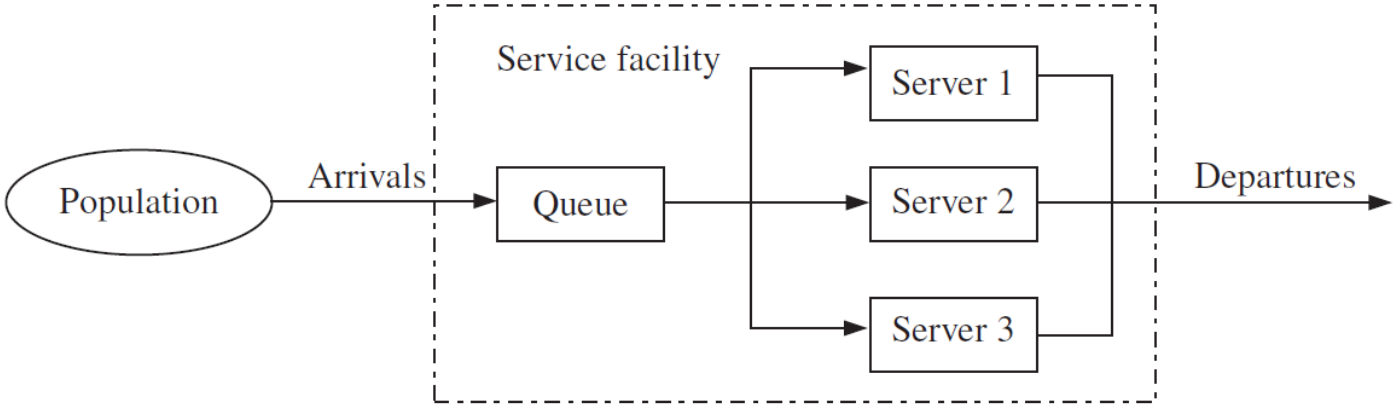
Introduction:

A queue is a waiting line of customers/jobs/items

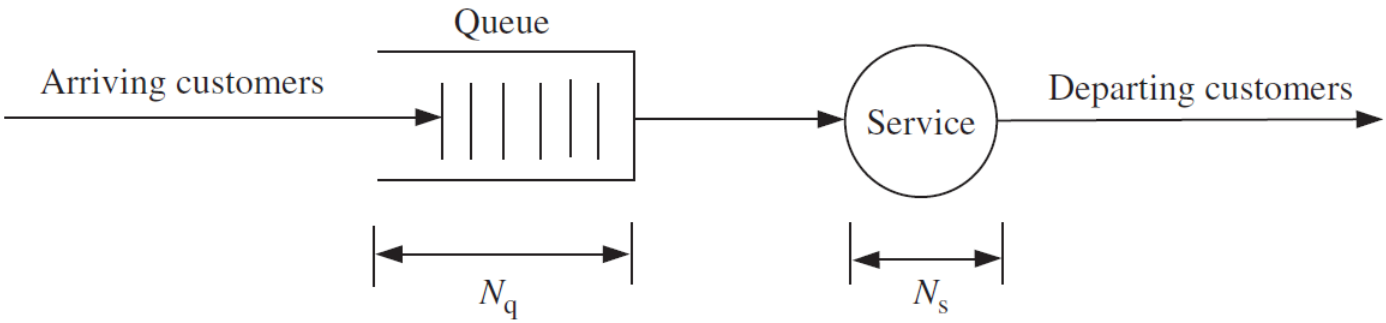
Examples:

Markovian Queueing systems

Components of a Queueing system



Queueing Process



Notations/Random Variables

a. N_q is the number of customers in queue.
b. N_s is the number of customers receiving service.
c. N is the total number of customers in the system: $N = N_q + N_s$.
d. W is the time a customer spends in queue before going to service, i.e., the waiting time.

e. X is the time a customer spends in actual service.
f. T is the total time a customer spends in the system (also called the sojourn time): $T = W + X$.

Components of a Queueing system

Kendal Notation

The Kendall notation is a shorthand notation that is used to describe queueing systems. It is written in the form:

$$A/B/c/D/E/F$$

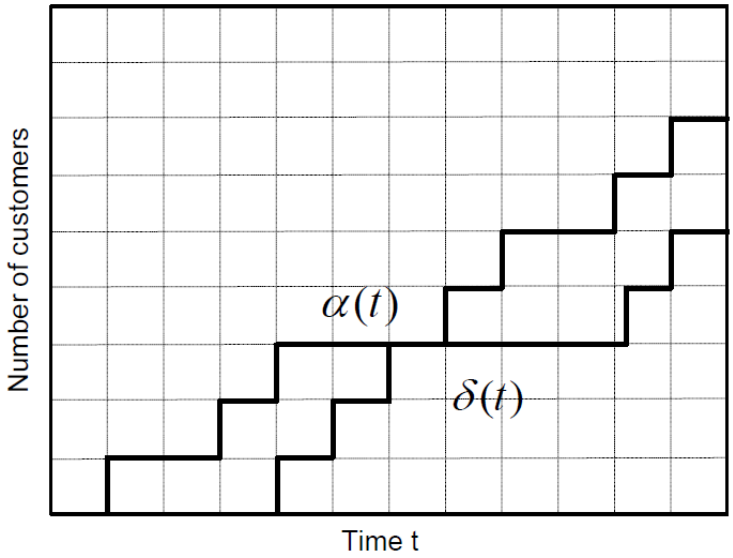
Symbols traditionally used for A and B are

Common Queueing systems

Little’s Formula

Establish a very simple relationship among

- Mean number in the system
- Mean arrival rate
- Mean system time



Area between two curves

Let us now assume that our queueing system is such that the following limits exist as $t \to \infty$.

$$\lambda = \lim_{t \to \infty} \lambda_t$$

$$T = \lim_{t \to \infty} T_t$$

Then, $\overline{N} = \lambda T$

This results is known as Little's result. It states that the average number of customers in a queueing system is equal to the average arrival rate of customers to that system, times the average time spent in that system.

M/M/1 Queueing System

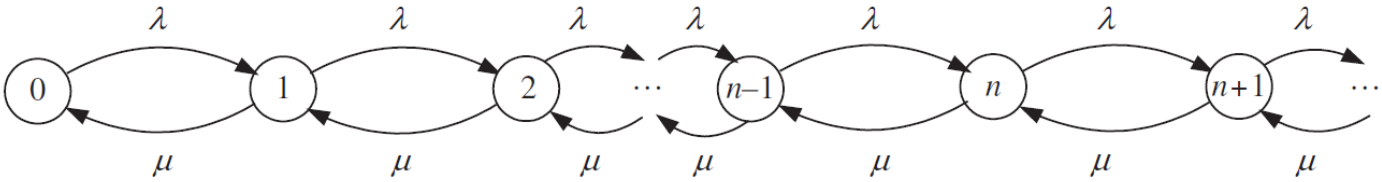


Figure 7.3 State transition rate diagram for M/M/1 queue.

$$P_n = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1\left(1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}\right)}, \quad n \geq 1$$

$$P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}}$$

$$P_n = P_0\left(\frac{\lambda}{\mu}\right)^n$$

$$P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\left(\frac{\lambda}{\mu}\right)^n}$$

$$P_0 = 1 - \frac{\lambda}{\mu}$$

Mean Number of customers in the system

Mean system time

Mean waiting time and queue length

Mean number of customers in service

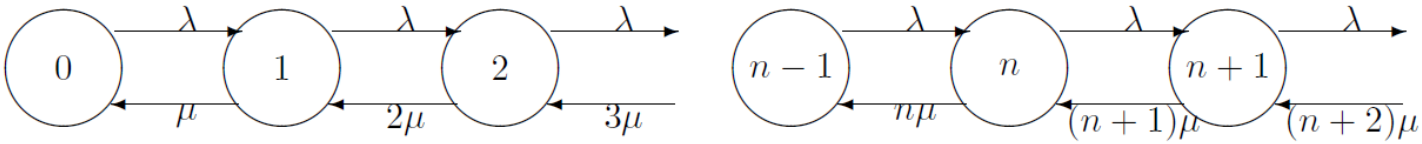
M/M/∞

In this system, we have

$$\lambda_n = \lambda, \quad n = 0, 1, 2, \cdots$$

$$\mu_n = n\mu, \quad n = 1, 2, \cdots$$

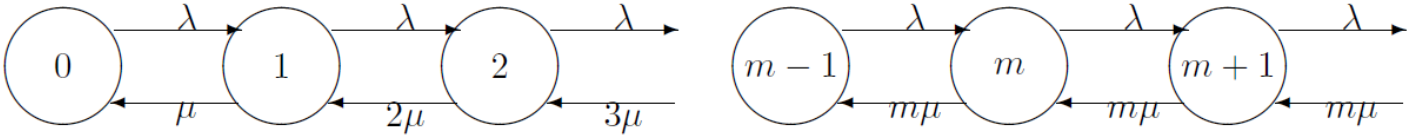
The state transition diagram is shown below:



$$P_n = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1\left(1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}\right)}, \quad n \geq 1$$

$M/M/m$ or $M/M/c$ Queueing Systems

The state transition diagram is shown below:



$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}} \qquad P_n = P_0\left(\frac{\lambda}{\mu}\right)^n$$

$$\sum_{n=0}^{\infty} p_n = 1 \Rightarrow p_0 = \frac{1}{\sum_{n=0}^{c-1}\left(\frac{\lambda}{\mu}\right)^n\left(\frac{1}{n!}\right) + \left(\frac{\lambda}{\mu}\right)^c\left(\frac{1}{c!}\right)\left(\frac{c\mu}{c\mu-\lambda}\right)}$$

The probability that an arriving customer has to wait, referred to as the delay probability, is

$$P_W = P[N \geq c] = P_c + P_{c+1} + P_{c+2} + \cdots$$

$$= \frac{P_c}{1-\frac{\lambda}{c\mu}} = \frac{P_c}{1-\rho}$$

$$= \frac{(c\rho)^c}{c!(1-\rho)]\ \sum_{n=0}^{c-1}\frac{(c\rho)^n}{n!}+(c\rho)^m}$$

The mean queue length is

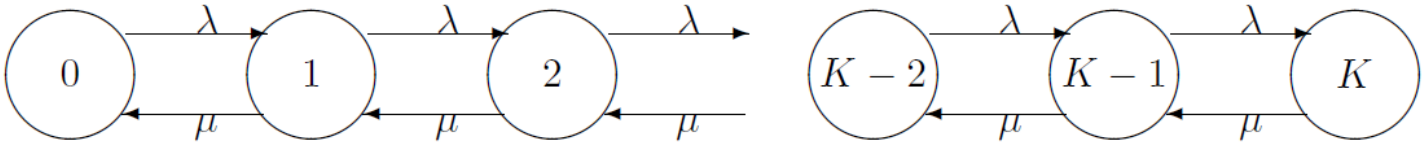
$$E[N_q] = \sum_{n=c}^{\infty} n p_n = p_c \sum_{n=0}^{\infty} n \rho^n = \frac{\rho p_c}{(1-\rho)^2} = \frac{\rho P_W}{1-\rho}$$

Using Little's formula, the mean waiting time

$$E[W] = \frac{E[N_q]}{\lambda} = \frac{p_c}{c\mu(1-\rho)^2} = \frac{P_W}{c\mu(1-\rho)}$$

$M/M/1/K$ Queueing Systems

The state transition diagram is shown below:



$$\lambda_n = \begin{cases} \lambda, & n < K \\ 0, & n \geq K \end{cases}$$

$$\mu_n = \mu, \quad n = 1, 2, \cdots, K$$

$$
\begin{aligned}
P_n &= P_0 \prod_{i=0}^{n-1} \frac{\lambda}{\mu} \\
&= P_0 \left(\frac{\lambda}{\mu}\right)^n, \quad n \leq K
\end{aligned}
$$

$$P_n = 0, \quad n > K$$

$$P_n = \begin{cases} \frac{1-\lambda/\mu}{1-(\lambda/\mu)^{K+1}} \left(\frac{\lambda}{\mu}\right)^n, & 0 \leq n \leq K \\ 0, & \text{Otherwise} \end{cases}$$

$$E[N] = \sum_{n=0}^{K} n p_n = p_0 \sum_{n=0}^{K} n \rho^n = \rho p_0 \frac{\mathrm{d}}{\mathrm{d}\rho} \left\{ \frac{1-\rho^{K+1}}{1-\rho} \right\}$$

$$= \rho p_0 \left\{ \frac{1-\rho^{K+1} - (1-\rho)(K+1)\rho^K}{(1-\rho)^2} \right\} = \rho p_0 \left\{ \frac{1-(K+1)\rho^K + K\rho^{K+1})}{(1-\rho)^2} \right\}$$

$$= \frac{\rho}{1-\rho^{K+1}} \left\{ \frac{1-(K+1)\rho^K + K\rho^{K+1})}{1-\rho} \right\} = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}}$$

$$E\left[N_{\mathrm{q}}\right] = E[N] - (1-p_0) = E[N] - \frac{\rho(1-\rho^K)}{1-\rho^{K+1}} = \frac{\rho}{1-\rho} - \frac{\rho(1+K\rho^K)}{1-\rho^{K+1}}$$

Customers are turned away with probability

$$p_K = \frac{(1-\rho)\rho^K}{1-\rho^{K+1}}$$

The actual rate at which customers arrive into the system, λ_A, as

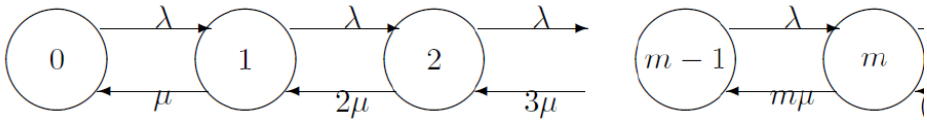
$$\lambda_A = \lambda(1-p_K) = \frac{\lambda(1-\rho^K)}{1-\rho^{K+1}}$$

We can then apply Little's formula to obtain

$$E[T] = \frac{E[N]}{\lambda_A} = \frac{1}{\mu}\left\{\frac{1}{1-\rho} - \frac{K\rho^K}{1-\rho^K}\right\}$$

$$E[W] = \frac{E[N_q]}{\lambda_A} = \frac{1}{\mu}\left\{\frac{\rho}{1-\rho} - \frac{K\rho^K}{1-\rho^K}\right\}$$

$M/M/m/m$ Queueing Systems



$$\lambda_n = \begin{cases} \lambda, & n < m \\ 0, & n \geq m \end{cases}$$

$$\mu_n = n\mu, \quad n = 1, 2, \cdots, m$$

$$P_n = P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu}, \quad n \leq m$$

$$P_n = \begin{cases} P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!}, & n \leq m \\ 0, & n > m \end{cases}$$

$$P_0 = \left(\sum_{k=0}^{m} \left(\frac{\lambda}{\mu}\right)^k \frac{1}{k!} \right)^{-1}$$

$$P_m = \frac{(\lambda/\mu)^m / m!}{\sum_{k=0}^{m} (\lambda/\mu)^k / k!}$$

This equation is called as Erlang B formula

Or Erlang's loss formula

The actual average arrival rate is $\lambda_A = \lambda(1 - P_m)$

Because no customer is allowed to wait, $E[W]$ and $E[N_q]$ are both zero.

Mean number of customers in the system is

$$E[N] = \sum_{n=0}^{c} np_n = p_0 \sum_{n=1}^{c} n \frac{(\lambda/\mu)^n}{n!} = (\lambda/\mu) p_0 \sum_{n=1}^{c} \frac{(\lambda/\mu)^{n-1}}{(n-1)!}$$

$$= (\lambda/\mu) p_0 \sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} = (\lambda/\mu)[1 - p_c]$$

By Little's formula,

$$E[T] = E[N]/\lambda_A = 1/\mu$$

Ex 5.3.2 Consider a cellular system where handoff calls are prioritized over originating calls by assigning a portion of channels exclusively for handoff calls. Let's assume that S_R channels are assigned exclusively for handoff calls among S channels in a cell. The remaining $S_C(= S - S_R)$ channels are shard by both calls. Find the blocking probability of originating call and handoff call. Assume call arrival and departure follows Poisson processes. And the arrival rates of originating and handoff calls are λ_o and λ_h, respectively, and the service rate is μ.

From the local balance equation,

$$n\mu P_n = (\lambda_0 + \lambda_h) P_{n-1} \quad 0 \leq n \leq S_c$$

$$n\mu P_n = \lambda_h P_{n-1} \quad S_c < n \leq S$$

$$P_n = \begin{cases} \frac{(\lambda_0+\lambda_h)^n}{\mu^n n!} P_0, & 0 \leq n \leq S_c \\ \frac{(\lambda_0+\lambda_h)^{S_c} \lambda_h^{n-S_c}}{\mu^n n!} P_0, & S_c < n \leq S \end{cases}$$

$$P_0 = \left(\sum_{i=0}^{S_c} \frac{(\lambda_0 + \lambda_h)^i}{\mu^i i!} + \sum_{i=S_c+1}^{S} \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{i-S_c}}{\mu^i i!} \right)^{-1}$$

Let P_b and P_d denote blocking probability and handoff dropping probability.

$$P_b = \sum_{n=S_c}^{S} P_n$$

$$P_d = P_S = \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{S-S_c}}{\mu^S S!} P_0$$